

Root Cause Analysis Playbook - Boiler Tube Leak

The single largest cause of forced outages in Indian coal units

Field	Value
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Equipment OEM	(synthetic OEM)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Why it matters

Boiler tube leaks (BTL) are the leading contributor to forced outages and lost availability in coal-fired units. Early detection and disciplined RCA materially improve EAF/EFOR.

2. Mechanisms (by location)

Mechanism	Typical location	Driver
Fireside erosion	Economizer, LTSH	Fly-ash velocity, high ash coal
Overheating / creep	SH/RH platens	Steam-side deposits, low flow
Corrosion fatigue	Waterwall	Water chemistry, cyclic stress
Soot-blower erosion	Near soot blowers	Misaligned/steam-wet blowing
Hydrogen damage	Waterwall	Under-deposit acidic corrosion

3. Early Indicators

- Rising furnace/tube-leak acoustic monitor readings.
- Make-up water / DM consumption increase.
- Unaccounted drop in boiler efficiency; flue-gas moisture rise.
- Local temperature anomalies on metal thermocouples.

4. Prevention & Response

- Maintain water/steam chemistry within limits; monitor deposits.
- Coal quality control (ash, abrasives); manage fly-ash velocity.
- Soot-blower alignment and steam-quality checks.
- On confirmed leak: plan controlled shutdown; isolate; pad-repair or tube replacement (typical MTTR 5 days).